java Program

Area and perimeter of a circle)

Q1 . Write a program that displays the area and perimeter of circle that has a radius of 5.5 using the following formula:

```
import java.util.Scanner ;

class Circle
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.println("Eneter a Radius (cm):");

        double radius = sc.nextDouble();

        final double PI =27/7.0;

        final double AREA = PI *(radius*radius);

        final double PERIMETER = 2 * PI * radius;

        System.out.println( "Radius :" + radius + "cm");


       System.out.printf("Area of Circle : %.3f cm ^ 2 \n" ,A
REA);
      System.out.printf("Perimeter of Circle : %.3f cm  \n" ,
PERIMETER);


    }
}
```

(Population projection)

The Indian Census Bureau projects population based on the following assumptions:

One birth every 7 seconds

One death every 13 seconds

One new immigrant every 45 seconds

Q. 2 Write a program to display the population for each of the next five years. Assume the current population is 312,032,486 and one year has 365 days.

```
class Population
{
    public static void main(String[] args)
    {

            long currPop = 312032486;

                        //  day  hr  min sec   year

               long seconds = (365 *24* 60 * 60) * 5 ;


            final Long BIRTH = seconds / 7;

            final Long DEATH = seconds/13;

            final Long  IMMIGRANTS =seconds /45;


            long newPop = currPop + BIRTH - DEATH + IMMIGRANT
S;


            System.out.println("Previous Pop :- " + currPop);

            System.out.println("New Pop : " + newPop);


    }
}
```

Average speed in miles)

Q. Assume a runner runs 14 kilometers in 45 minutes and 30 seconds. Write a program that displays the average speed in miles per hour.

(Note that 1 mile is 1.6 kilometers.)

```
class AverageMiles
{
    public static void main(String[] args)
    {
```

```
      double miles = 14.0 / 1.6;

       double hour = (45.0/60)+(30.0/3600);


       double Speed = miles / hour;


        System.out.printf("Average Speed : %.3f mph%n", Spee
d);

    }
}
```

Convert Celsius to Fahrenheit)

Q. Write a program that reads a Celsius degree in a double value from the console, then converts it to Fahrenheit and displays the result.

The formula for the conversion is as follows:

Fahrenheit = (9/5) * celsius + 32

Hint: In Java, 9/5 is 1, but 9.0/5 is 1.8.2

OUTPUT: Enter a degree in Celsius: 43

43 Celsius is 109.4 Fahrenheit

```

import java.util.Scanner;

class CelsiusFahrenheit
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a degree in Celsius:");

        double celsius =sc.nextDouble();

          double fahrenheit = (9.0/5) * celsius+32;


         System.out.printf("%.0f Celsius is  %.1f Fahrenheit%
n", celsius, fahrenheit);

    }
}
```

(Compute the volume of a cylinder)

Q. Write a program that reads in the radius and length of a cylinder and computes the area and volume using the following formulas:

area 3.14 radius * radius

volume = area * height

Here is a sample run:

2 Enter the radius and height of a cylinder: 5.5   12

The area is 95.0331

The volume is 1140.4

```
import java.util.Scanner;


class Cylinder
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a radius:-");
        double radius = sc.nextDouble();

        System.out.print("Enter a height:-");
        double height = sc.nextDouble();

        final double PI = 3.14159;

        final double  AREA =PI * radius * radius;

        final double VOLUME = AREA* height;

        System.out.printf("The Area is:- %.4f \n" , AREA);

      System.out.printf("The Volume is:- %.1f \n" , VOLUME);

    }
}
```

Convert feet into meters)

Q. Write a program that reads a number in feet, converts it to meters, and displays the result. One foot is 0.305 meter.

Here is a sample run:

Enter a value for feet: 16.5

16.5 feet is 5.0325 meters

```
import java.util.Scanner;

class FeetToMeter
{
    public static void main(String[] args)
    {

       Scanner sc =new Scanner(System.in);

       System.out.println("Enter a value for feet:");

       double feet = sc.nextDouble();


       double meter = feet * 0.305;

        System.out.printf("%.1f Feet is %.4f  meters \n ",fee
t,meter);
    }
}
```

Convert pounds into kilograms)

Q. Write a program that converts pounds into kilograms. The program prompts the user to enter a number in pounds, converts it to kilograms, and displays the result. One pound is 0.454 kilograms.

Here is a sample run:

Enter a number in pounds: 55.5 55.5 pounds is 25.197 kilograms

```
import java.util.Scanner;

class PoundsToKg
{
    public static void main(String[] args)
    {
        Scanner sc =new Scanner(System.in);

        System.out.println("Enter a  number in pounds:");

        double pound = sc.nextDouble();

        double kg = pound * 0.454;

        System.out.printf("%.1f Pounds is %.3f Kilograms" ,po
```

```
und,kg);
    }
}
```